

CHIRAG HARIPRASAD

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Summary

AI/ML engineer with experience in real-time financial systems, document intelligence, and fraud detection infrastructure (IIFL Samasta Finance). 4 patents published (V.E.C.T.O.R, G.E.N.E.S.I.S, O.R.I.O.N, PresntAI); published in IEEE Access (S.C.A.L.E.). Led the team behind 3 systems built during a fintech internship. Sub-50ms streaming ML pipelines.

Education

RV College of Engineering (RVCE), Bengaluru

July 2024 – June 2027

B.E. in Artificial Intelligence and Machine Learning CGPA: 8.23/10.0

Sri Jayachamarajendra (Govt.) Polytechnic (SJP), Bengaluru

Aug 2021 – May 2024

Diploma in Mechatronics Engineering CGPA: 9.64/10.0 (Top 1% of Class)

Internship Experience

IIFL Samasta Finance Ltd.

Bengaluru, Karnataka

Strategy Team - AI Intern

Nov 2025 – May 2026 (6 Months)

- Led a team building **3 systems** spanning full-stack, ML, and API infrastructure.
- VerifAI (KYC Document Verification)**: Vision-language document classification using the Gemini API with structured prompting and downgrade-first validation logic; multi-class output (LEGIT/SUSPICIOUS/FRAUD/DAMAGED) deployed in the KYC workflow for live loan applications, with a full audit trail (CSV/JSON) for RBI compliance.
- MeetingsAI (On-Prem Meeting Intelligence)**: Offline pipeline with in-UI recording, speaker diarization (pyannote), speaker recognition (SpeechBrain ECAPA), transcription (faster-whisper), and multi-model LLM analysis via Ollama; auto-emails MoM and transcripts to stakeholders, with no cloud dependency.
- Lifestyle Index (B2B API-as-a-Service)**: FastAPI gateway with API key auth, background scoring jobs, rate limiting, and analytics; exposes risk scores to external partners through admin dashboards and score-card UI.

Selected Technical Projects

V.E.C.T.O.R Velocity-Enhanced Clustering for Transactional Outlier Recognition (*Patent Published*)

2025

- Real-time fraud detection engine personalizing anomaly detection per user via UMAP + HDBSCAN behavioral clustering, per-cluster Isolation Forest models, and an XGBoost cold-start fallback; scores transactions in **sub-50ms**. Patent No. 202541077457.
- Streaming pipeline on Redis Streams with MongoDB persistence and a live React dashboard hosted via Cloudflare Tunnel.

S.C.A.L.E. Structural Causal Analysis of Labor & Education (*Published, IEEE Access*)

2025 – 2026

- Three-layer simulation engine (System Dynamics ODEs + Structural Causal Model with do-operator interventions + Agent-Based Model) producing 15-year policy projections for India's engineering education system.
- Quantified ATE of **+44.8 pp** graduate unemployment per +1 pp enrollment share, calibrated on national datasets (AICTE, AISHE, PLFS, RBI) spanning 2010–2023.

G.E.N.E.S.I.S Generative Engine for Networked Spatial Infrastructure Synthesis (*Patent Published*)

2026

- Converts site boundaries into NBC-compliant city layouts (roads, blocks, plots) with a fail-fast regulation engine enforcing FAR, setbacks, and land-use rules inside the generation loop.
- Runs traffic (graph routing), drainage (D8 flow), and wind (Gaussian wake) proxy simulations per layout, feeding a weighted KPI scoring and Top-K ranking engine.

Key Achievements

Patents: 4 patents published (2025–26) V.E.C.T.O.R, G.E.N.E.S.I.S, O.R.I.O.N, PresntAI.

Research: S.C.A.L.E. published in **IEEE Access**

Academic: Ranked **2nd in class**, CGPA **9.64/10.0** in Diploma; current RVCE CGPA **8.23/10.0**.

Leadership: Class Representative **3 consecutive years** (60+ peers); led RILM Project Sahakarita v2.0 deploying 1000+ systems for 10,000+ students across 3 schools.

Technical Skills

Languages: Python, JavaScript/TypeScript, C.

ML / AI: PyTorch, TensorFlow, scikit-learn, XGBoost, FAISS, HDBSCAN.

Backend: FastAPI, Flask, REST APIs, Redis Streams, JWT/RBAC.

Frontend: React, TailwindCSS, Vite.

Cloud & Infra: AWS (Lambda, SQS, CloudFront), Docker, MongoDB, PostgreSQL, Linux.

AI Platforms: Gemini API, Groq API, Ollama, Hugging Face.